MILITARY SPECIFICATION

HOUSING, ATOMIC WEAPON, ADAPTION KIT, SUBASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of atomic weapon control support entitled Housing, Atomic Weapon, Adaption Kit, Subassembly which is used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48776	- Adaption Kit, Sub-assembly, Atomic Weapon Ml.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8808362	- Container, Packing, Ammunition, for Domestic (Conus) Shipments of Special Weapons Items, Packing and Marking.
8813453	- Control Support Cover.
8822046	- Carton Packing Ammunition for Support Atomic Weapon Control.
8826452	- Housing, Atomic Weapon, Adaption Kit, Subassembly.

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8826454 - Adaption Kit, Subassembly, Atomic Weapon: M1.
8826455 - Terminal Board Assembly, Connectors.
EL-8842557 - Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The adaption kit subassembly atomic weapon housing shall comply with all requirements specified on the drawing (dwg) 8826452, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Circuit resistance before and after operation. The resistance of each circuit shall not exceed the value stated for the respective circuit in Table I when determined as specified in 4.4.1.

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Table I

Support circuit resistance

Item No.	Between Terminals	Maximum Resistance ohms, 160 + 5°F 80 + 20°F	Maximum Resistance ohms, -65 + 5°F
1	1J1-B 1J1-E	0.020	0.016
2	1J1-D Ground	0.020	0.016
3	1J1-H 1J10-9	0.020	0.016
4	1J1-J Ground	0.020	0.016
5	1J2-A 1J2-T	0.020	0.016
6	1J2-D Ground	0.020	0.016
7	1J2-E Ground	0.020	0.016
8	1J2-F 1J2-V	0.020	0.016
9	1J2-J Ground	0.020	0.016
10	1J2-K Ground	0.020	0.016
11	1J2-P 1J1-A	0.020	0.016
12	1J2-U Ground	0.020	0.016
13	1J2-W Ground	0.020	0.016
14	1J2-Y 1J4-55	0.025	0.020
15	1J2-a Ground	0.020	0.016
16	1J2-$\overline{b}$ Ground	0.020	0.016
17	1J3-$\overline{A}$ 1J3-K	0.020	0.016
18	1J3-B 1J3-E	0.020	0.016
19	1J3-D Ground	0.025	0.020
20	1J3-L Ground	0.025	0.020
21	1J4-1 1J5-L	0.035	0.029
22	1J4-2 1J7-6	0.025	0.020
23	1J4-4 1P1-W	0.040	0.033
24	1J4-6 1J7-6	0.025	0.020
25	1J4-7 1J7-6	0.025	0.020
26	1J4-13 1J7-3	0.025	0.020
27	1J4-14 Ground	0.030	0.024
28	1J4-17 Ground	0.030	0.024
29	1J4-21 1J4-9	0.020	0.016
30	1J4-21 1P2-R	0.020	0.016
31	1J4-25 1J4-10	0.020	0.016
32	1J4-27 1J4-47	0.025	0.020
33	1J4-27 1J5-B	0.020	0.016
34	1J4-31 1P2-M	0.020	0.020
35	1J4-31 1P2-e	0.025	0.020
36	1J4-31 1J4-$\overline{52}$	0.025	0.020
37	1J4-33 1P2-B	0.020	0.016

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Item No.	Between Terminals		Maximum Resistance ohms, 160 + 5°F 80 + 20°F	Maximum Resistance ohms, -65 + 5°F
38	1J4-33	1J4-32	0.025	0.020
39	1J4-33	1P2-K	0.020	0.016
40	1J4-34	1J4-16	0.025	0.020
41	1J4-34	1J4-15	0.030	0.024
42	1J4-34	1J4-30	0.030	0.024
43	1J4-36	1J6-35	0.030	0.024
44	1J4-41	Ground	0.030	0.024
45	1J4-42	Ground	0.030	0.024
46	1J4-43	Ground	0.030	0.024
47	1J4-44	Ground	0.030	0.024
48	1J4-54	1J2-X	0.030	0.024
*49	1J4-54	Ground	0.33	0.285
*50	1J4-55	Ground	0.33	0.285
51	1J4-56	1J4-57	0.025	0.020
52	1J4-58	1J4-38	0.025	0.020
53	1J4-58	1J2-C	0.030	0.024
54	1J4-58	1J2-H	0.030	0.024
55	1J4-60	1J4-12	0.020	0.016
56	1J4-60	1P2-U	0.020	0.016
57	1J4-60	1P2-V	0.020	0.016
58	1J4-64	1J5-E	0.025	0.020
59	1J5-C	1J5-F	0.020	0.016
60	1J5-C	1J4-18	0.025	0.020
61	1J5-C	1J3-M	0.020	0.016
62	1J5-C	1J1-F	0.020	0.016
63	1J5-C	1J3-J	0.020	0.016
64	1J5-C	1J1-N	0.020	0.016
65	1J5-C	1J3-H	0.020	0.016
66	1J5-C	1J1-L	0.020	0.016
67	1J5-C	1J3-F	0.020	0.016
68	1J5-C	1J1-K	0.020	0.016
69	1J5-C	1J1-B	0.020	0.016
70	1J5-C	1J1-E	0.020	0.016
71	1J5-C	1J3-E	0.020	0.016
72	1J5-C	1J3-B	0.020	0.016
73	1J5-H	1P1-Y	0.030	0.024
74	1J5-J	1J4-57	0.030	0.024
75	1J5-K	1J6-6	0.035	0.029
76	1J6-1	Ground	0.020	0.016

*Items 49 and 50 - A direct current (dc) power resistance tester must be used.

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Item No.	Between Terminals		Maximum Resistance ohms, 160 + 5°F 80 + 20°F	Maximum Resistance ohms, -65 + 5°F
77	1J6-3	1J6-22	0.020	0.016
78	1J6-3	1P1-K	0.020	0.016
80	1J6-8	1J6-27	0.020	0.016
81	1J6-9	1J6-28	0.020	0.016
82	1J6-9	1P2-1	0.025	0.020
83	1J6-11	Ground	0.020	0.016
84	1J6-14	1J6-32	0.020	0.016
85	1J6-14	1P2-L	0.025	0.020
86	1J6-15	1J6-33	0.020	0.016
87	1J6-18	1J6-19	0.020	0.016
88	1J6-18	1J4-37	0.030	0.024
89	1J6-21	1J4-19	0.040	0.033
92	1J6-25	1P1-E	0.015	0.012
93	1J6-29	Ground	0.020	0.016
94	1J6-30	Ground	0.020	0.016
95	1J6-35	1J6-36	0.020	0.016
96	1J7-7	Ground	0.030	0.024
97	1J8-A	1J8-B	0.020	0.016
98	1J8-C	1J10-2	0.035	0.029
99	1J8-D	1J11-19	0.030	0.024
100	1J8-E	1J11-15	0.030	0.024
101	1J8-F	1J11-1	0.035	0.029
102	1J8-G	1J11-5	0.030	0.024
103	1J8-H	1J10-5	0.035	0.029
104	1J8-J	1P1-X	0.030	0.024
105	1J8-K	1P1-S	0.030	0.024
106	1J8-L	1J10-23	0.025	0.020
107	1J8-M	1J10-10	0.035	0.029
108	1J8-N	1J10-24	0.035	0.029
109	1J8-P	1J10-25	0.035	0.029
110	1J8-R	1J11-25	0.025	0.020
111	1J8-S	1J11-24	0.025	0.020
112	1J8-T	1J11-23	0.025	0.020
113	1J8-U	1J11-10	0.025	0.020
114	1J8-V	1J9-9	0.025	0.020
115	1J8-W	1J9-5	0.025	0.020
116	1J8-X	1P1-V	0.030	0.024
117	1J8-Y	1J3-G	0.025	0.020
118	1J8-Z	1P2-a	0.050	0.042
119	1J8-a	1P2-F	0.050	0.042

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Item No.	Between Terminals		Maximum Resistance ohms, 160 + 5°F 80 + 20°F	Maximum Resistance ohms, -65 + 5°F
120	1J8-b	1J10-15	0.035	0.029
121	1J8-c	1J10-19	0.035	0.029
122	1J8-d	1J5-B	0.030	0.024
123	1J8-e	1P2-Y	0.050	0.042
124	1J8-f	1P2-S	0.050	0.042
125	1J8-g	1J1-G	0.025	0.020
126	1J8-h	1J1-I	0.025	0.020
127	1J8-i	1J1-M	0.025	0.020
128	1J8-k	1J1-C	0.025	0.020
129	1J8-m	1J3-C	0.030	0.024
130	1J8-n	1P1-F	0.025	0.020
131	1J8-p	1P1-U	0.030	0.024
132	1J8-q	1P1-H	0.025	0.020
133	1J8-r	1J7-5	0.030	0.024
134	1J8-s	1J7-4	0.030	0.024
135	1J8-t	1J1-H	0.035	0.029
136	1J8-u	1J1-A	0.025	0.020
137	1J8-v	1P2-g	0.050	0.042
138	1J8-w	1P2-H	0.050	0.042
139	1J8-x	1J7-2	0.030	0.024
140	1J8-y	1P1-N	0.025	0.020
141	1J8-z	1J6-2	0.020	0.016
142	1J8-AA	1J6-17	0.020	0.016
143	1J8-BB	1J6-37	0.020	0.016
144	1J8-CC	1J6-18	0.045	0.037
145	1J8-DD	1J6-35	0.045	0.037
146	1J8-EE	Ground	0.020	0.016
147	1J8-FF	1P1-C	0.025	0.020
148	1J8-GG	1J4-57	0.040	0.033
149	1J8-HH	1J7-8	0.030	0.024
150	1J9-1	1P2-H	0.040	0.033
151	1J9-2	1J9-11	0.020	0.016
157	1J9-12	1J4-34	0.035	0.029
158	1J9-14	1J9-15	0.020	0.016
159	1J9-14	1J4-51	0.025	0.020
160	1J10-1	1J2-A	0.020	0.016
161	1J10-1	1J10-2	0.020	0.016
162	1J10-6	1P1-Z	0.035	0.029
163	1J10-8	1J10-3	0.020	0.016
164	1J10-12	1J9-2	0.020	0.016
165	1J10-12	1J10-13	0.020	0.016
166	1J10-14	1J11-16	0.030	0.024

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Item No.	Between Terminals	Maximum Resistance ohms, 160 + 5°F 80 + 20°F	Maximum Resistance ohms, -65 + 5°F
167	1J10-17 Ground	0.020	0.016
168	1J10-18 Ground	0.020	0.016
169	1J10-21 Ground	0.020	0.016
170	1J10-22 Ground	0.020	0.016
171	1J11-1 1J11-2	0.020	0.016
172	1J11-1 1J2-F	0.020	0.016
173	1J11-3 1J11-8	0.020	0.016
174	1J11-4 1J10-4	0.035	0.029
175	1J11-6 1P1-Z	0.025	0.020
176	1J11-9 1J7-1	0.025	0.020
177	1J11-12 1J11-13	0.020	0.016
178	1J11-12 1J9-2	0.020	0.016
179	1J11-14 1J10-16	0.030	0.024
180	1J11-17 Ground	0.025	0.020
181	1J11-18 Ground	0.025	0.020
182	1J11-21 Ground	0.025	0.020
183	1J11-22 Ground	0.025	0.020
188	1P1-A 1J9-8	0.020	0.016
189	1P1-B 1P1-D	0.020	0.016
190	1P1-B 1J5-A	0.020	0.016
191	1P1-J Ground	0.020	0.016
192	1P1-K 1P1-M	0.020	0.016
193	1P1-L Ground	0.020	0.016
194	1P1-P 1J11-11	0.025	0.020
195	1P1-P 1J10-11	0.030	0.024
196	1P1-R 1P1-T	0.020	0.016
197	1P1-R 1J9-14	0.020	0.016
198	1P1-AA 1J6-7	0.020	0.016
199	1P1-BB 1J6-13	0.020	0.016
200	1P1-CC 1J6-26	0.020	0.016
201	1P1-DD 1J6-31	0.020	0.016
202	1P2-C 1P2-g	0.035	0.029
203	1P2-D 1J6-8	0.025	0.020
204	1P2-G 1J6-15	0.025	0.020

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Item No.	Between Terminals	Maximum Resistance ohms, 160 + 5°F 80 + 20°F	Maximum Resistance ohms, -65 + 5°F	Test Code No.
205	1P2-N Ground 1R2	370 Min - 430 Max	370 Min - 430 Max	030001
206	1P2-0 Ground	0.025	0.020	
207	1P2-P 1J4-25	0.025	0.020	
208	1P2-T Ground	0.025	0.020	
209	1P2-b 1J2-L	0.045	0.037	
210	1P2-c Ground 1R1	370 Min - 430 Max	370 Min - 430 Max	030002
211	1P2-k 1J2-M	0.045	0.037	
212	1P2-j 1J2-N	0.045	0.037	

Note: In order to check the resistance of a specific connection, other than 1J8-EE to ground, a four wire test circuit shall be used with one current and potential connection made to the specific connection. The ground connection shall be made by connecting the second potential lead to 1J8-EE and the second current lead to the support housing. In order to check the resistance of 1J8-EE to "ground", a current and potential lead shall be connected to 1J8-EE. The second current lead shall be connected to the supporting housing and the second potential lead shall be connected to any pin connector that is already connected to ground.

3.7 Insulation resistance. The insulation resistance between each terminal and all other terminals specified in Table II shall not be less than 20 megohms at -65 degrees Fahrenheit (°F) plus or minus (±) 5°F or less than 1 megohm at 160 ± 5°F and 80 ± 20°F when tested as specified in 4.4.2.

Table II

Connector terminals

1J8-A	1J8-W	1J8-r	1P2-D	1J6-6	1J9-8
1J8-C	1J8-X	1J8-s	1P2-I	1J6-7	1J9-12
1J8-D	1J8-Y	1J8-t	1P2-M	1J6-13	1J9-14
1J8-E	1J8-Z	1J8-u	1P2-P	1J6-14	1J10-3
1J8-F	1J8-a	1J8-v	1P2-R	1J6-15	1J10-7
1J8-G	1J8-b	1J8-w	1P2-U	1J6-21	1J11-3
1J8-H	1J8-c	1J8-x	1J2-C	1J6-23	1J11-4
1J8-J	1J8-d	1J8-y	1J2-L	1J6-24	1J11-6
1J8-K	1J8-e	1J8-z	1J2-M	1J6-25	1J11-7
1J8-L	1J8-f	1J8-AA	1J2-N	1J6-26	1J11-9
1J8-M	1J8-g	1J8-BB	1J3-A	1J6-31	1J11-11
1J8-N	1J8-h	1J8-CC	1J5-A	1J7-3	1J11-12
1J8-P	1J8-i	1J8-DD	1J5-E	1J7-6	1J11-14
1J8-R	1J8-k	1J8-FF	1J5-F	1J9-3	1J11-16
1J8-S	1J8-m	~~1J8~~-GG	1J5-H	1J9-4	1P1-W
1J8-T	1J8-n	1J8-HH	1J5-L	1J9-6	
1J8-U	1J8-p		1J6-3	1J9-7	*Unused Pins
1J8-V	1J8-q		1J6-5		

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*1J2- R,S,Z,c; 1J3-1,N; 1J4-3,5,8,11,20,22,23,24,26,28,29,35, 39,40,45,46,48,49,50,53,59,61,62,63; 1J5-D,G; 1J6-4,10,12,16, 20,34; 1J7-9,10,11,12,13,14,15; 1J9-13; 1J10-20; 1J11-20.

Note: All these unused terminals may be connected together and checked as one terminal.

3.8 Pulse circuit operation. A voltage pulse as specified by Figure 1 (generated by the circuit of Figure 2 or equivalent) shall be applied to pins B and E of 1J2. The output pulse, measured at pins 55 and 41 of 1J4 across a 2.4 ohms ± 1 percent resistive load shall be 6.0 volts minimum (min) for a duration of 10 microseconds and shall exceed 3.0 volts for 110 microseconds min. The above test shall be repeated with the input pulse applied to pins G and E of 1J2 and output monitored at pins 54 and 41 of 1J4. The test shall be conducted as specified in 4.4.3.

Pulse transformer output connector pins	Specified pulse width seconds	Test Code No.
1J2-B to 1J2-Y	0.00015 minimum	050001
1J2-G to 1J2-X	0.00015 minimum	050002

3.9 Voltage regulation. The zener diode voltage drops shall meet the requirements specified in Table III. Test 3 and 4 of Table III shall be limited to 5.0 seconds maximum (max). The test shall be conducted as specified in 4.4.4.

Table III

Test	Load current Milliamperes	Test connections From +	To -	Zener voltage volts	Test Code No.
1	70 + 5	1P2-B	1P2-c	2.8 Min	060001
2	70 + 5	1P2-B	1P2-$\overline{N}$	2.8 Min	060002
3	500 - 25	1P2-B	1P2-c	4.50 Max	060003
4	500 - 25	1P2-B	1P2-$\overline{N}$	4.50 Max	060004

3.10 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall conducted in the order specified.

3.10.1 High temperature. The assembly shall withstand a high temperature of 160 ± 5°F without evidence of damage and shall meet the requirements of 3.6 through 3.9 while at 160 ± 5°F when tested as specified in 4.4.5.6.

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3.10.2 Low temperature. The assembly shall withstand a low temperature of -80 ± 5°F without evidence of damage and shall meet the requirements of 3.6 through 3.9 while at -65 ± 5°F when tested as specified in 4.4.5.2.

3.10.3 Transportation vibration. The control support shall be vibrated at the frequencies, accelerations, and temperatures for the time specified in Table IV without evidence of damage such as cracks, chips, separations, or loose parts. The vibration shall be applied in each of three mutually perpendicular axes, one of which is in the direction of forward motion (see Figure 3), and shall comply with 3.6 through 3.9 at 80 ± 20°F, after being tested as specified in 4.4.5.3.

Table IV

Frequency range Hertz	Double amplitude or peak g level ± 10% tol.	Cycle time to cover frequency range (minutes)	Total test time in minutes per axis (+ 10% tol.) 80 + 20°F	160 + 5°F	-65 + 5°F
10-20-10	± 2 g	3 ± 0.5	15	10	10
20-50-20	0.036 inch	3 ± 0.5	15	10	10
50-100-50	± 5.0 g	3 ± 0.5	9	6	6
100-500-100	± 1 g	3 ± 0.5	6	3	3

3.10.4 Flight vibration. The assembly shall withstand subjection to vibration over a frequency range of 75 to 1000 to 75 Hertz (Hz) at a peak level of 5 gravity units (g's) ± 10 percent. The vibration shall be applied from a low to a high frequency and returned to a low frequency over a period of 0.9 minutes ± 0.1 minutes in the axis along the direction of forward motion (see Figure 3). The monitor circuit specified in Table V shall show no evidence of open circuit during test when tested as specified in 4.4.5.4. Upon completion of this test the assembly shall comply with 3.6 through 3.9.

Table V

Monitor circuit connections	Terminals connected together
1J8-d to 1J8-EE	1J1-A to 1J1-D 1J3-D to 1J3-E 1J5-C to 1J5-B 1J2-W to 1J2-P

3.10.5 Transportation shock. The assembly shall withstand shocks of 75 ± 8 g's peak amplitude with a half amplitude duration of 9 ± 3 milliseconds without evidence of damage when tested as specified in 4.4.5.5. Upon completion of this test the assembly shall comply with 3.6 through 3.9.

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3.10.6 Impact shock. The assembly shall withstand a shock of 150 ± 15 g's peak with a half amplitude duration of 4 ± 1 milliseconds applied once in the axial direction opposite that of forward motion (see Figure 3). The monitor circuit specified in Table V shall show no evidence of open circuit when tested as specified in 4.4.5.6. Upon completion of this test the assembly shall comply with 3.6 through 3.9.

3.11 Soldering. All soldering shall comply with the requirements of MIL-STD-1460.

3.12 Test codes (see 3.5). The contractor shall provide data feedback on punch cards in conformance with Quality Assurance Instructions QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.13.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 4 complete assemblies (see dwg 8822046) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

Table VI

First article inspection and tests

Inspection or Test	Unit Number	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	1,2,3 and 4		X
2. Circuit resistance (see 3.6)	1,2,3 and 4		X
3. Insulation resistance (see 3.7)	1,2,3 and 4		X
4. Pulse circuit operation (see 3.8)	1,2,3 and 4		X
5. Voltage regulator (see 3.9)	1,2,3 and 4		X
6. High temperature (see 3.10.1)	1,2,3 and 4		X
7. Low temperature (see 3.10.2)	1,2,3 and 4		X
8. Transportation vibration (see 3.10.3)	1	X	
9. Flight vibration (see 3.10.4)	2	X	
10. Transportation shock (see 3.10.5)	3	X	
11. Impact shock (see 3.10.6)	4	X	

*If the producer who is supplying this Housing is also supplying the Adaption Kit Sub-Assembly, Atomic Weapon: M1 per MIL-A-48776 then the initial production testing per paragraphs 3.9, 3.10, 3.11, 4.4.7, 4.4.8, 4.4.9, 3.10.3, 3.10.4, 3.10.5, 4.4.5.3, 4.4.5.4 and 4.4.5.5 of this specification may be deferred since similar tests are being performed on the next assembly in accordance with MIL-A-48776.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Terminal board assembly, connectors (see dwg 8826455 covering a detail of dwg 8826452).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Portion of connector broken away	Visual
	102. Improper soldering (see 3.11)	Visual
	103. Evidence of poor workmanship (see 3.13)	Visual
	104. Mechanically improper wiring arrangements (wires in tension, excess wire lengths, inadequate or damaged insulation)	Visual/Manual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Housing, atomic weapon, adaption kit subassembly prior to potting (see dwg 8826452).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101. Any misaligned, loosely connected or mislocated component parts or assemblies.....Visual
102. Evidence of poor workmanship (see 3.13).....Visual
103. Required screws, washers, spacers, nuts, insulator, terminal board assemblies or control support missing.....Visual
104. Excess strain on leadwires of connectors that are assembled directly to support housing.....Visual
105. Soldering improper (see 3.11).....Visual
106. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:

201. Serialization improper or missing (see 3.4).....Visual

4.3.2.1.3 Housing, atomic weapon, adaption kit subassembly (see dwg 8826452).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper location of 4 mounting holes, 2 aligning pins and connector for battery assembly, XM69	Gage
102.	Improper location of 3 mounting holes and connector for integrating accelerometer	Gage
103.	Improper location of 2 mounting holes and connector for acceloswitch	Gage
104.	Improper location of 4 mounting holes and connector for timer	Gage
105.	Improper relationship of 5 mounting holes, for Test Set, Atomic Weapon, VT 4133	Gage
106.	Improper relationship of 5 mounting holes for Switch, Rotary Fuze, Option XM14	Gage

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107.	Improper location of 9 clearance holes and 8 tapped holes in housing flange	Gage
108.	Improper location of 4 tapped holes and 1 clearance hole for receptacle in housing leg	Gage
109.	Improper male terminal location (J1 and J2 conn.)	Gage
110.	Improper location of signal light holes	Gage
111.	Improper amount of potting material	Visual and Depth Micrometer
112.	Wrong connector	Visual
*113.	Height of pins (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
**114.	Depth of sockets (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual
115.	Deformation of connector sufficient to impair mating	Visual
116.	Permanently damaged terminals (gouged, nicked or flattened)	Visual
117.	Any pin missing (as evidenced by empty hole in insert)	Visual
118.	Cracks, voids, or bubbles in the area potted or wire raised out of the potting which would interfere with or be abraded by the components	Visual
119.	Any pin bent	Visual
120.	Any misaligned, loosely connected or mislocated component parts or assemblies	Visual
121.	Evidence of poor workmanship (see 3.13)	Visual

Minor: None defined.

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*In the event pins do not appear to be within 1/16 inch in height relative to the adjacent pins, a gage (see 4.3.4.1) shall be used to determine that variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to adjacent sockets, a gage (see 4.3.4.1) shall be used to determine that variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector shall be utilized as a reference surface for all measurements of socket depths.

4.3.2.1.4 Carton, packing, ammunition for support, atomic weapon, control (see dwg 8822046).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Wrong size bag	Visual
102.	Bag not sealed	Visual
103.	Air in bag not minimized before sealing	Visual
104.	Marking improper, illegible, smeared, missing, or in wrong location	Visual
105.	Packing material visibly wet	Visual
106.	Incorrect or insufficient cushioning material	Visual
107.	Evidence of poor workmanship (see 3.13)	Visual
Minor:	None defined.	

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4.3.2.1.5 Container packing ammunition for domestic (CONUS) shipments of special weapons items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Marking improper, illegible, smeared, missing or in wrong location	Visual
102.	Packing material visibly wet	Visual
103.	Incorrect or insufficient cushioning material	Visual
104.	Evidence of poor workmanship (see 3.13)	Visual
Minor:	None defined.	

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Circuit resistance	Major
b.	Insulation resistance	Major
c.	Pulse circuit operation	Major
d.	Voltage regulation	Major
e.	Transportation vibration	Major
f.	Flight vibration	Major

Of these tests, (e) and (f) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit resistance, insulation resistance, pulse circuit operation and voltage operation. Each assembly shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.6 through 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration and flight vibration. A sample shall be selected at random in accordance with Table IX. The sample shall be tested in accordance with 4.4.5.3 and 4.4.5.4. If any item fails to comply with 3.10.3 and 3.10.4, the lot shall be rejected.

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Table IX

Sample sizes for destructive testing

Number required to be Delivered	Sample Size for No Failures
3-500	3
501-800	5
801-1300	7

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842557). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The resistance of the circuits specified in Table I shall be measured with a suitable four terminal instrument capable of measuring the values specified.

4.4.2 Insulation resistance. An insulation resistance tester, capable of supplying the required voltage of 500 $\pm$ 50 volts direct current (vdc), and with which a resistance in megohms can be measured shall be used to measure the resistance. The current shall be limited to a maximum of 5 milliamperes to protect human life.

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4.4.3 Pulse circuit operation. Pulse generating and monitoring equipment shall be connected to connector pins of 1J2 and 1J4 as shown on Figure 2. The pulse generator (Figure 2 or equivalent) shall have the characteristics specified in Figure 1 and shall be operated for each pulse transformer. (The pulse generator of Figure 2 is operated by momentarily throwing switch S1 to position 2 then returning to position 1.) The monitoring equipment shall consist of instruments capable of measuring a transient peak voltage of 6.0 volts for a duration of 10 microseconds and the time interval of an output pulse level of more than 3.0 volts for 110 microseconds min.

4.4.4 Voltage regulation. Current generator circuits capable of developing currents of 70 ± 5 and 500 + 0-25 milliamperes through the zener diode circuits specified in Table III shall be used. The voltage shall be monitored at output terminals specified in Table III with an instrument capable of measuring 0 to 4.50 volts. Tests 3 and 4 of Table III shall be limited to 5 seconds max. per test. Tests may be repeated providing a minimum test "off" time of 30 seconds follows the 5 second max. test "on" time.

4.4.5 Environmental testing (see 3.10).

4.4.5.1 High temperature. The assembly shall be placed in the test chamber and the temperature of chamber raised to 160 ± 5°F at a relative humidity of not more than 5 percent. The assembly shall soak at this temperature for a minimum of 4 hours. During this 4 hour period the assembly shall be tested as specified in 4.4.1 through 4.4.4.

4.4.5.2 Low temperature. The assembly shall be protected from frost and placed in the test chamber and the temperature of the chamber lowered to -80 ± 5°F. The item shall soak at this temperature for a minimum of 4 hours. The chamber temperature shall then be raised to -65 ± 5°F and maintained for for an additional 1 hour period or until stabilization (see 6.4) is reached, whichever is longer. The assembly shall be tested at -65 ± 5°F as specified in 4.4.1 through 4.4.4.

4.4.5.3 Transportation vibration. The assembly shall be rigidly mounted to a fixture capable of being supported in each of three mutually perpendicular axes with one of these axes along the direction of forward motion of the assembly, and the fixture mounted on a suitable vibration producing apparatus. The assemlby shall be vibrated at the frequencies, amplitudes, temperatures, and for the time specified in Table IV, in each of three mutually perpendicular axis. Before starting this test, the assembly cover (8813453) or equivalent must be assembled to the assembly as indicated on dwg 8826454. Protect from frost during cold temperatures. Following this test, the assembly shall be tested as specified in 4.4.1 through 4.4.4.

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4.4.5.4 Flight vibration. The assembly shall be rigidly mounted to a fixture capable of supporting the assembly in the direction of forward motion, and the fixture mounted on a suitable vibration producing apparatus. The assembly shall be vibrated over the frequecy, amplitudes, temperature, and time specified. The monitor circuit in Table V shall be monitored throughout the test for evidence of closure. Before starting this test, the assembly cover (8813453) or equivalent must be assembled to the assembly as indicated on dwg 8826454. Upon completion of this test the assembly shall be tested in accordance with 4.4.1 through 4.4.4.

4.4.5.5 Transportation shock. The assembly shall be rigidly mounted to a fixture capable of being supported in each of three mutually perpendicular axes, with one of these axes along its direction of forward motion (see Figure 3), and the fixture mounted to a suitable shock producing device. The assembly shall be subjected to shock three times in each direction in each of three mutually perpendicular axes for a total of 18 shocks. Before starting this test, the assembly cover (8813453) or equivalent must be assembled to the assembly as indicated on dwg 8826454. Following this test, the assembly shall be tested as specified in 4.4.1 though 4.4.4.

4.4.5.6 Impact shock. The assembly shall be rigidly mounted to a fixture capable of being supported in the forward axial direction. The assembly shall be subjected to shock. The circuit specified in Table V shall be monitored throughout the test to insure continuity with not more than 20 microseconds interruption. Before starting this test, the assembly cover (8813453) or equivalent must be assembled as indicated on dwg 8826454. Upon completion of this test the assembly shall comply with 4.4.1 through 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg 8822046 using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg 8822046 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The assembly shall be packed in accordance dwg 8822046.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822046.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.6, 3.8, 3.9 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1675A dated 24 January 1967.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A165

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